BRAIN TISSUE BINDING BY EQUILIBRIUM DIALYSIS METHOD

Protocol Number :	APS-DMPK-402-NVS	Effective Date:	01 Jul 2022
Version :	2.0	Review Date:	30 Jun 2024

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
%	Percentage
°C	Degree Celsius
µL	Microlitre
µM	Micromolar
COA	Certificate of analysis
conc^n	Concentration
DMPK	Drug metabolism and pharmakokinetics
DMSO	Dimethyl Sulfoxide
eg	For example
g	Gram
HCl	Hydrochloric acid
HOD	Head of Department
HPLC	High Performance Liquid Chromatography
hr	Hour
ID	Identification
LC-MS/MS	Liquid chromatography coupled with tandem mass spectrometry
LNB	Lab note book
LOQ	Limit of Quantification
M	Molarity
min	Minutes
mM	Millimolar
MS	Microsoft
N	Normality
Na_2HPO_4	Disodium hydrogen phosphate
KPO_4	Potassium phosphate
NaOH	Sodium hydroxide
KH_2PO_4	Potassium phosphate monobasic
nM	Nanomolar
pH	Negative logarithm to base 10 of the positive hydrogen ion molar concentration
PPE	Personal protective equipment
RED	Rapid equilibrium dialysis
rpm	Revolutions per minute
S.No.	Serial number

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1. OBJECTIVE

To study the brain tissue binding potential of test compound(s) following equilibrium dialysis method.

2. SCOPE

The procedure described in this protocol will be used for the assessment of brain tissue binding of test compound(s) in rat brain homogenate.

3. RESPONSIBILITY

All personnel performing this study are responsible to follow the process outlined in this protocol, at Aurigene Pharmaceutical Services Limited, India.

4. INTRODUCTION

Estimation of the unbound fraction of drug in brain provides a valuable tool for identifying new therapies for central nervous system (CNS) diseases. The extent of partitioning into brain tissue influences CNS penetration which in turn influences the efficacy and / or toxicological effects of a drug. Although the brain receives about one-sixth of cardiac output, drug penetration to brain cannot be measured by plasma protein binding assay. It is because the composition of brain and plasma are very different; therefore free drug fraction in plasma cannot replace unbound brain concentrations.

The main reason for different drug penetration characters is blood-brain barrier (BBB). BBB is a highly selective semipermeable membrane that separates blood circling from CNS and the brain extracellular fluid. It consists of the endothelium of brain capillaries and astrocyte. The endothelial cells of brain capillaries are more tightly joined than most capillaries in other tissues, resulting in slower passive diffusion of water-soluble drugs. Furthermore, it selectively transports of lipophilic compounds by way of active transport. The astrocytic sheath consists of a layer of glial connective tissue cells (astrocytes) close to the basement membrane of the capillary endothelium and plays an important role in secretion and absorption of neural transmitters. The binding of drugs to proteins both in the vascular space and/ or the extravascular space results in a decrease in drug clearance and a prolonged drug half-life. Only the unbound drug is available for glomerular filtration and in some cases, hepatic clearance.

Equilibrium dialysis is an accurate and reliable method for determining brain tissue binding affinity of chemical or biological substances of low molecular weights.

Rat brain homogenate

Aliquot	5 mL
Gender	mixed
Homogenate media	100mM KPO_4
Homogenate Ratio	1:3Tissue(g):Homogenate media
Maximum freeze/thaw cycles	1
Storage	-80°C
Strain	Sprague Dawley
Supplier	In-house

5. SAFETY AND PRECAUTIONS

During the conduct of the study, the study personnel should take necessary safety precautions and use personal protective equipment (PPE) such as face masks, gloves, lab coat, goggles, shoes etc.

DMSO and acetonitrile:

- Keep away from sources of ignition.
- Do not breathe vapour or mist.
- Avoid contact with skin, eyes and clothing.

6. MATERIALS AND EQUIPMENT

6.1 EQUIPMENT

Equipment	Make
Adhesive permeable seals	Thermo Scientific
Analytical balance	Sartorius/ Denver
Centrifuge with plate/ tube rotors	Eppendorf/ Kubota & Thermo
CO_2 Incubator	Shalom instruments
Cyclomixer	IKA®Vortex
Deep freezer (-80°C)	Thermo Scientific/ Sanyo
Deep well plate (96-well)	Axygen scientific
Incubator	SHEL Labs
LC-MS/MS (API-5500, API- 4000, API- 3200)	MDS/ AB Sciex
Micro centrifuge tubes	Tarsons/ Phenomenex
pH meter	Horiba scientific
Pipettes (2.5, 10, 100, 200 and 1000 µL)	Eppendorf
RED device Inserts	Thermo Scientific
Refrigerator	LG
Shaker (37°C)	Heidolph
Software	Microsoft excel
Sonicator	Shalom instruments
Teflon® base plate	Thermo Scientific
Water bath	Julabo corporation

Note:

- Details of all equipment/ instruments used during the study should be provided in the LNB & report (if any).

6.2 CHEMICALS

Chemical	Make; Catalogue No
Acetonitrile, HPLC grade	J.T.Baker; 9012-03
Dimethyl sulfoxide	Sigma-Aldrich; D8418
Ethanol	Changshu hong sheng fine chemicals Cooperation, Limited.
Glyburide	Sigma Aldrich; G2539
Potassium Chloride	Sigma; 87526
Potassium phosphate monobasic monohydrate	Sigma Aldrich; P5655
Propranolol	Sigma-Aldrich; 82067
Sodium Chloride	Sigma; 71386
Sodium phosphate dibasic heptahydrate	Sigma Aldrich; 30413

Note:

- Complete list of chemicals used and their details should be provided in the LNB & report (If any).
- Equivalent grade chemicals, reagents or alternate instruments of different source/ make can be used on need basis.

6.3 PREPARATION OF SOLUTIONS AND STORAGE CONDITIONS

S. No.	Composition	Storage condition
1.	Preparation of test/ reference compound(s) Prepare 10 mM stock solution in DMSO. Further dilute the stock in DMSO to obtain 0.5 mM solution and subsequently in brain to obtain a 5 µM working solution maintaining the organic solvent ≤ 1%.	RT Working solution: Prepare fresh
2.	Homogenate media and /or dialysis buffer preparation Add 23 g sodium phosphate dibasic and 2 g of potassium phosphate monobasic in 1000 mL of de-ionized water. Add 668 µL of 4000 mM potassium chloride solution and 27.59 mL of 5000 nM sodium chloride solution to above solution. Check and adjust pH to 7.4.	2-8°C Use within one month
3.	Preparation of 20% v/v ethanol Add 200 mL of ethanol in 800 mL of de-ionized water (v/v) to obtain 20% ethanol.	Prepare fresh
4.	Preparation of stop solution Prepare 1 mg/mL stock solution of Glyburide (internal standard) and dilute with acetonitrile to obtain appropriate concentration.	2-8°C

Note:

- Changes to the above mentioned solution method preparation should be captured in the LNB & report (If any).

7. PROCEDURE

7.1 PREPARATION OF TEFLON® BASE PLATE

Rinse the Teflon® base plate wells with 20% ethanol for 10 min.

Remove alcohol and rinse twice with de-ionized water.

Allow the plate to dry for few min.

7.2 EQUILIBRIUM DIALYSIS

The assay is to be carried out in triplicates (n=3).

Pre-warm the dialysis buffer (100 mM KPO_4, pH 7.4) in an incubator at 37°C.

Thaw the rat brain homogenate at 37°C

Prepare the test and reference compounds as described in section 7.3.

For recovery, draw 50 µL brain homogenate samples from the working solution prior to incubation (T_0 hr) and dispense in to separate wells in a deep well plate (96-well).Add 50 µL of buffer and mix well to normalize the matrix.

Place inserts (open end up) into the wells of the Teflon® base plate.

Add 300 µL of working solution (brain homogenate) to the brain chamber indicated by the red ring and 500 µL of dialysis buffer to the buffer chamber.

Seal the unit with plate sealer and incubate at 37°C on an orbital shaker at 750 rpm, 5% CO_2 for 4 hr.

Remove plate sealer and take out 50 µL from both buffer and brain homogenate chambers and dispense in to separate wells in a 96 well-deep well plate.

Add 50 µL of blank brain homogenate to the buffer samples and 50 µL of buffer to the brain homogenate samples and mix well to normalize the matrix.

To the above wells add 300 µL of stop solution and centrifuge the samples for 15 min at 4000 rpm at 4°C.

Take out the supernatant, transfer into LC-MS/MS vials and submit for analysis.

Discard RED inserts, wash the Teflon® base plate with 20% ethanol, dry and keep for reuse.

Note:

- Modifications to the procedure, if any, should be captured in the LNB & report.
- Appropriate bio-analytical method will be used for the quantification of the analyte

8. ASSAY CONTROL

Propranolol should be included as assay control.

Study would be qualified based on the acceptance criteria given in the data template.

Note:

- Any changes to the assay controls would be made only with the consent of the sponsor and should be described in the LNB & report ((If any).

9. DATA RECORDING

The data should be tabulated in MS excel based brain tissue binding assay template and the final data should be pasted in the LNB. Raw data should be recorded in the LC-MS/MS server with the study number and the data path should be provided in the excel file and LNB.

10. DATA ANALYSIS

Peak area ratio (test compound peak area/ internal standard peak area) in the brain homogenate and buffer samples can be used to determine percent bound. Determine recovery of the test compound in brain homogenate using (T_0 hr) sample. Calculate the percentage bound, percentage unbound and percentage recovery by using the formula given below.

The fraction unbound measured in the homogenate fu-hom is calculated as follows:

$$\text{fu-hom} = \frac{\text{PAR buffer (post incubation)}}{\text{PAR homogenate (post incubation)}}$$

The fraction unbound for non-diluted tissue fu-tissue is calculated as :

$$\text{fu-tissue} = \frac{1/D_f}{(1/f_{u\text{-hom}}-1) + 1/D_f} = \frac{f_{u\text{-hom}}}{D_f - (D_f - 1)f_{u\text{-hom}}}$$

where

fu-hom is the fraction unbound in the homogenate and D_f is the dilution factor of the homogenate

(D_f = 4 according the our protocol)

The % free is calculated as : % free = 100 x fu-tissue

The %bound is calculated as : % bound = 100 - % free

A recovery value is calculated using the equation below:

$$\%\ \text{Recovery} = \frac{\text{PAR buffer (post-incubation)* Vol buffer chamber + PAR brain (post-incubation) * Vol brain chamber}}{\text{PAR brain (pre-incubation) * Vol brain chamber}} \times 100$$

$$\text{LOQ \%fu} = \frac{\text{3x Blank}}{\text{Average } (T_0 \text{ area plasma})} \times 100$$

11. REPORTING

Data should be reported in an MS excel format.

For the experiment to be published, the data for Propranolol should be within assay limits, the % recovery and % stability should be >70%.

Condition	Report Status	Remark
Precipitation observed in tube	No data reported	Precipitation in DMSO stock solution
No peak present in plasma at time=0	Undefined	Test compound not detectable in samples
Plasma time=0 peak area < lower limit	No data reported	Analytical issue - no MS/too low signal
Poor peak shape observed and accuracy of result compromised	No data reported	Analytical issue – Poor chromatogram
No compound available for assay	Undefined	Empty tube
% free <1%	Report as <1%	N/A
% free >80% (very highly unbound compounds)	Report % free as >80%	N/A
Recovery between <55% or >130%	No data reported	Recovery outside of acceptance criteria
Recovery more than 80%	Report as >=80	N/A
CV(%free n=3) >50%	if CV(% free n=2) <50% , report % free calculated with 2 replicates	% free calculated with only 2 replicates
CV(%free n=3) >50%	if CV(% free n=2) >50%, do not report any results for %free	Undefined - High variability in % free
% free >1 % AND % free < LOQ % free	No data reported	LOQ issue

12. DATA ARCHIVAL

All the raw data, chromatograms and reports should be stored in the server at Aurigene Pharmaceutical Services Limited, India for a period of two years or as specified by sponsor. Assay details and analytical procedures should be captured in LNB and same should also be archived as per the instructions/ procedure of Intellectual Property Management (IPM) department.

13. REFERENCES

- http://www.cambridgemedchemconsulting.com/resources/ADME/distribution.html.
- Wright JD, Boudinot FD, Ujhelyi MR; Measurement and analysis of unbound drug concentrations; Clinical Pharmacokinetics; Jun; 30 (6); 445-62; 1996.

14. CHANGE AND AMENDMENT HISTORY

Change history		
Version No.	**Changes made/ Reason for changes**	**Effective date**
2.0	Revised Version	01 Jul 2022